THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 19	Substitute both values.
2	**(c)** $4x + 7y$	Collect the like terms separately.
3	**(a)** 23	Work out the bracket first.
4	**(b)** $5x + 7$	Subtract the x-terms and the constants separately.
5	**(b)** 43	Substitute the value of n.
6	**(b)** $3x + 5y$	Multiply each price by its quantity.
7	**(a)** 15	$a^2 = a \times a$.
8	**(d)** 11	Work out the numerator, then divide.
9	**(b)** 30	Substitute both values.
10	**(c)** $8x + 9y$	Collect the like terms separately.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Expand the bracket: $3x + 12$ (1); Add $5x$: $8x + 12$ (1).	1+1
12	$(9x + 2) - (3x - 5)$ (1); Answer: $6x + 7$ (1).	1+1
13	Substitute: $5^2 + 3(5) - 4$ (1); $= 36$ (1).	1+1
14	Expand the bracket: $2x + 14$ (1); Add $6x$: $8x + 14$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 10n + 3$ (1); Difference $= 4n + 7$ (1); At $n = 4$: sum $= 43$ and difference $= 23$ (1).	1+1+1
16	For 9 triangles: $2(9) + 1 = 19$ (1); For 14 triangles: $2(14) + 1 = 29$ (2).	1+2
17	Fare $= 45 + 11k$ (1); For 8 km: $45 + 11(8) = 133$ (1); For 13 km: $45 + 11(13) = 188$ (1).	1+1+1
	OR: Sum $= 14n + 3$ (1); Difference $= 4n + 5$ (1); At $n = 3$: sum $= 45$ and difference $= 17$ (1).	1+1+1
18	For 12 triangles: $2(12) + 1 = 25$ (1); For 20 triangles: $2(20) + 1 = 41$ (2).	1+2
	OR: Fare $= 60 + 9k$ (1); For 12 km: $60 + 9(12) = 168$ (1); For 17 km: $60 + 9(17) = 213$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sales $= 48x + 36y$ (1); At the given values: $480 + 540$ (2); Total sales $=$ ₹1020 (2).	1+2+2
	OR: Mother now $= 3r + 4$ (1); After 5 years: Raju $= r + 5$, mother $= 3r + 9$ (1); Sum $= 4r + 14$ (1); At $r = 10$: 54 years (2).	1+1+1+2
20	Sales $= 45x + 30y$ (1); At the given values: $900 + 300$ (2); Total sales $=$ ₹1200 (2).	1+2+2
	OR: Mother now $= 3r + 6$ (1); After 3 years: Raju $= r + 3$, mother $= 3r + 9$ (1); Sum $= 4r + 12$ (1); At $r = 12$: 60 years (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 31	Work out the bracket first.
2	**(b)** $6x + 6$	Subtract the x-terms and the constants separately.
3	**(b)** 73	Substitute the value of n.
4	**(d)** $6x + 2y$	Multiply each price by its quantity.
5	**(b)** 36	$a^2 = a \times a$.
6	**(d)** 9	Work out the numerator, then divide.
7	**(a)** 28	Substitute both values.
8	**(c)** $5x + 11y$	Collect the like terms separately.
9	**(b)** 27	Work out the bracket first.
10	**(c)** $7x + 3$	Subtract the x-terms and the constants separately.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(11x + 6) - (4x - 1)$ (1); Answer: $7x + 7$ (1).	1+1
12	Substitute: $6^2 + 2(6) - 9$ (1); $= 39$ (1).	1+1
13	Expand the bracket: $4x + 12$ (1); Add $2x$: $6x + 12$ (1).	1+1
14	$(8x + 3) - (2x - 7)$ (1); Answer: $6x + 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 10n + 3$ (1); Difference $= 6n + 9$ (1); At $n = 5$: sum $= 53$ and difference $= 39$ (1).	1+1+1
16	For 15 triangles: $2(15) + 1 = 31$ (1); For 25 triangles: $2(25) + 1 = 51$ (2).	1+2
17	Fare $= 35 + 14k$ (1); For 6 km: $35 + 14(6) = 119$ (1); For 11 km: $35 + 14(11) = 189$ (1).	1+1+1
	OR: Sum $= 16n - 1$ (1); Difference $= 4n + 7$ (1); At $n = 2$: sum $= 31$ and difference $= 15$ (1).	1+1+1
18	For 7 triangles: $2(7) + 1 = 15$ (1); For 18 triangles: $2(18) + 1 = 37$ (2).	1+2
	OR: Fare $= 50 + 13k$ (1); For 10 km: $50 + 13(10) = 180$ (1); For 15 km: $50 + 13(15) = 245$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sales $= 38x + 28y$ (1); At the given values: $608 + 672$ (2); Total sales $=$ ₹1280 (2).	1+2+2
	OR: Mother now $= 3r + 2$ (1); After 8 years: Raju $= r + 8$, mother $= 3r + 10$ (1); Sum $= 4r + 18$ (1); At $r = 9$: 54 years (2).	1+1+1+2
20	Sales $= 52x + 40y$ (1); At the given values: $624 + 320$ (2); Total sales $=$ ₹944 (2).	1+2+2
	OR: Mother now $= 3r + 5$ (1); After 6 years: Raju $= r + 6$, mother $= 3r + 11$ (1); Sum $= 4r + 17$ (1); At $r = 11$: 61 years (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 47	Substitute the value of n.
2	**(c)** $4x + 7y$	Multiply each price by its quantity.
3	**(c)** 40	$a^2 = a \times a$.
4	**(a)** 7	Work out the numerator, then divide.
5	**(a)** 33	Substitute both values.
6	**(c)** $6x + 10y$	Collect the like terms separately.
7	**(b)** 21	Work out the bracket first.
8	**(c)** $5x + 9$	Subtract the x-terms and the constants separately.
9	**(a)** 54	Substitute the value of n.
10	**(c)** $8x + 3y$	Multiply each price by its quantity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $4^2 + 5(4) - 7$ (1); $= 29$ (1).	1+1
12	Expand the bracket: $5x + 10$ (1); Add $8x$: $13x + 10$ (1).	1+1
13	$(12x + 9) - (5x - 4)$ (1); Answer: $7x + 13$ (1).	1+1
14	Substitute: $7^2 + 4(7) - 3$ (1); $= 74$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 17n + 1$ (1); Difference $= 7n + 13$ (1); At $n = 3$: sum $= 52$ and difference $= 34$ (1).	1+1+1
16	For 20 triangles: $2(20) + 1 = 41$ (1); For 30 triangles: $2(30) + 1 = 61$ (2).	1+2
17	Fare $= 40 + 16k$ (1); For 7 km: $40 + 16(7) = 152$ (1); For 12 km: $40 + 16(12) = 232$ (1).	1+1+1
	OR: Sum $= 15n + 1$ (1); Difference $= 7n + 3$ (1); At $n = 6$: sum $= 91$ and difference $= 45$ (1).	1+1+1
18	For 11 triangles: $2(11) + 1 = 23$ (1); For 16 triangles: $2(16) + 1 = 33$ (2).	1+2
	OR: Fare $= 55 + 12k$ (1); For 9 km: $55 + 12(9) = 163$ (1); For 14 km: $55 + 12(14) = 223$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sales $= 60x + 35y$ (1); At the given values: $540 + 490$ (2); Total sales $=$ ₹1030 (2).	1+2+2
	OR: Mother now $= 3r + 3$ (1); After 4 years: Raju $= r + 4$, mother $= 3r + 7$ (1); Sum $= 4r + 11$ (1); At $r = 14$: 67 years (2).	1+1+1+2
20	Sales $= 44x + 32y$ (1); At the given values: $660 + 640$ (2); Total sales $=$ ₹1300 (2).	1+2+2
	OR: Mother now $= 3r + 7$ (1); After 2 years: Raju $= r + 2$, mother $= 3r + 9$ (1); Sum $= 4r + 11$ (1); At $r = 13$: 63 years (2).	1+1+1+2

Mistakes to avoid:

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- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
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